

S&P; SENTINEL

Financial-Text Risk Intelligence & Wholesale Portfolio Stress Testing Platform

Candidate: Aman Gupta | **Institution:** Indian Institute of Technology Kharagpur (IIT Kharagpur)
Event: S&P; Global & CRISIL Campus Hackathon 2026 | **Track:** Risk Intelligence Track
Localhost Policy: 100% Offline Runtime • Zero Cloud Keys • SHA-256 Audited Datasets

The Institutional Challenge	The Sentinel Solution
Traditional institutional risk monitoring operates with latency. By the time corporate earnings reports or debt rating downgrades are published, counterparty contagion and market spreads have already moved against the bank balance sheet.	Sentinel provides an auditable, sub-150ms risk intelligence engine. It extracts grounded financial signals from unstructured news and social streams, runs multi-hop contagion propagation, and revalues wholesale portfolio positions in real time.

System Architecture & Localhost Execution

End-to-End Pipeline & Strict Security Compliance

- **Multi-Source Ingestion:** Structured news CSVs, noisy social chatter with cashtags (\$APEX, \$TSTEL), reference universe of 23 tickers, and Kaggle open benchmarks.
- **Replay Controller & Deduplication:** Clock-controlled logical replay (1x-60x) with token Jaccard (≥ 0.65) and containment (≥ 0.80) near-duplicate suppression.
- **Hybrid NLP Risk Engine:** CPU-optimized FinBERT sentiment distribution, 10-class financial event classifier, exact character span grounding, and 1-10 severity rubric.
- **Contagion Knowledge Graph:** NetworkX directed customer-supplier and creditor-debtor graph with 2-hop dampening (0.50/hop) to prevent runaway feedback.
- **Wholesale Stress Valuation:** Real-time mark-to-market and Expected Credit Loss (ECL) revaluation across loans, corporate bonds, SOFR swaps, and retail credit.
- **Cryptographic Provenance:** 15 datasets tracked in data/manifest.json with SHA-256 checksums. Verified zero external cloud API calls in backend runtime.

Multi-Source NLP Risk Intelligence

Entity Linking, Sentiment Scoring, and Severity Rubric

Entity Linking & Exact Spans	Domain Sentiment Scoring
• Links corporate mentions to canonical tickers via alias tables. • Extracts exact start/end character offsets for textual evidence grounding. • Handles ambiguous references and cashtag tokens seamlessly.	• FinBERT & Lexicon hybrid calibrated to financial vernacular. • Strict probability distribution: $P(\text{pos}) + P(\text{neg}) + P(\text{neu}) = 1.0$. • Continuous polarity score mapped to $[-1.0, +1.0]$ domain.
10-Class Event Taxonomy	Additive 1-10 Severity Rubric
• Structured classes: CREDIT, MACRO, SUPPLY_CHAIN, REGULATORY, EARNINGS, M&A;, CYBER, ESG, PRODUCT, OTHER. • Confidence thresholding with explicit fallback to OTHER to avoid false alarms.	• Standardized impact severity: Base Severity + Scope + Multipliers. • Dynamic calibration based on regulatory penalties and capital spend. • Severity MAE: 0.89 points against gold rubric (Target ≤ 1.50 pts).

Contagion Propagation & Deduplication

Network Topologies and Second-Order Risk Transmission

- **Rolling Window Deduplication:** Identical or near-duplicate news items within a rolling 24-hour window are suppressed using Jaccard word set similarity (≥ 0.65) and containment (≥ 0.80), eliminating duplicate shocks to balance sheets.
- **Directed Knowledge Graph:** Explicit corporate supply-chain linkages (e.g., APEX supplies TSTEL at 0.45 weight; TSTEL supplies VAUTO at 0.65 weight).
- **Geometric Multi-Hop Damping:** Contagion shock dampens across network hops (factor = $\text{weight} \times 0.50^{\text{hop}}$). Prevents artificial systemic runaway.
- **Second-Order Transmission Example:** A credit default shock at APEX propagates 1 hop to Titan Steel (TSTEL) and 2 hops to Vanguard Auto (VAUTO), while unaffected sectors remain insulated (0 bps change).

Module B — Wholesale Stress Testing

\$500M Institutional Book + \$50M Retail Credit Sleeve

Asset Class	Book Exposure	Valuation Metric	Stress Transmission Mechanism
Syndicated Corporate Loans	\$220,000,000	Expected Credit Loss (ECL)	$\Delta PD = \text{Baseline PD} \times \text{severity} \times \text{shock factor}$; $ECL = EAD \times \Delta PD \times LGD$
Corporate Fixed Bonds	\$200,000,000	Mark-to-Market (MtM)	$\Delta P = -\text{ModDuration} \times \Delta \text{Spread} + 0.5 \times \text{Convexity} \times (\Delta \text{Spread})^2$
SOFR Interest Rate Swaps	\$150,000,000 gross	Signed DV01 Sensitivity	$\Delta \text{MtM} = \text{Signed DV01} \times \Delta \text{Curve bps}$ (directional for pay vs receive)
Consumer/SME Credit Sleeve	\$50,000,000	Pooled ECL Tranches	Aggregated from Kaggle retail debt data mapped to 8 sectors
Systemic Macro Curve Shifts	Portfolio-wide	Yield Curve Shift ($\pm$ bps)	Applies systemic interest rate shocks directly across all asset sleeves

Quantitative Evaluation & Honest Benchmarks

105 De-Leaked Holdout Samples with Confusion Matrix & Baselines

Benchmark Metric	Measured Value	PRD Target	Baseline Comparison	Audit Verification
Entity Linking Precision	100.0%	≥ 90.0%	Keyword Baseline: 75.0%	Exact character spans, zero placeholders
Event Classification Macro-F1	0.382	≥ 0.70	Keyword Baseline: 0.448	High precision (~1.0), selective recall
Sentiment Macro-F1	0.340	≥ 0.75	Lexicon Baseline: 0.651	Continuous score MAE: 0.419 pts
Severity Rubric MAE	0.89 pts	≤ 1.50 pts	Constant Mean: 2.10 pts	90.5% within ±1.0 point of gold rubric
Adversarial Sample Accuracy	91.7%	≥ 80.0%	Adversarial Disambiguation	Correctly rejects rumors and denials

Reproducibility Guarantee: All metrics evaluated locally without internet access. Fully reproducible via: `uv run python scripts/run_evaluation.py`

Compliance, Integrity & Production Readiness

Zero-Data-Loss Standards and Institutional Governance

Hackathon Guidelines Compliance	Production S&P; / CRISIL Fit
• Section 8 (Confidentiality): Zero confidential client or proprietary rating data utilized. • Section 9 (IP & Licensing): MIT License for project-authored code, explicit attributions for CC BY-NC-SA and CC0 datasets. • Zero Secrets: verify_hygiene.py strictly validates no credentials or tokens exist.	• Feed Integration: Drop-in microservice exposing FastAPI endpoints (/api/signals, /api/stress/run). • Real-Time Audit Sink: data/signals.jsonl append-only log for regulatory compliance. • Zero Cloud Key Overhead: Runs entirely on standard commodity CPU hardware.